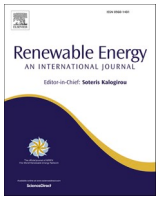
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# An efficient method for predicting PV modules performance based on the two-diode model and adaptable to the single-diode model

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ABSTRACT

This paper introduces a simplified and accurate modeling approach to model photovoltaic modules. The current method uses a hybrid technique combining numerical and analytical approaches to predict the current-voltage characteristics of PV generators by calculating the parameters of the double-diode model. First, to reduce the complexity of the current equation and minimize the calculation time, some simplifications reducing the research space from seven to only five unknowns are taken into consideration. Second, three of these five parameters are extracted using simple analytical equations derived from the output current's equation based on the available values of the typical points on the manufacturer's datasheet. Then, the remaining two parameters are extracted using a combination of numerical approach and slope adjustment technique at the short-circuit point. The technique is adapted to the single-diode model in order to study the effect of the second diode on the simulating accuracy. The effectiveness is tested over six PV modules matching different PV technologies, and the results are compared with a great number of modeling methods based on different approaches, and the adjustment technique has shown the highest accuracy levels (RMSE values less than 0.045A) at the least computational times (compilation time less than 0.084 s).

## 1. Introduction

To meet the growing energy requirements implied by today's technological development, nowadays, from an energy point of view, fossil fuels remain the most effective form of energy. And that makes them the most exploited source [1]. However, the massive widespread use of this last kind of energy has a considerable number of adverse effects. Burning fossil fuels causes the greenhouse effect by dropping large quantities of glasshouse gas into the atmosphere. And that directly implies the earth's temperature rising and the pollution of the air leading to several health diseases [2]. Moreover, the world's reserves of fossil fuels are limited and concentrated only in some countries. Thus, their overexploitation has caused a rapid decrease in their quantities leading to economic problems, especially for non-energy-producing countries [3]. In the last few decades, energies such as; solar energy, wind energy, and biomass energy, have emerged and are considered the alternatives that combine availability, cleanness, safety, and renewability [4].

The renewable energy revolution of today highlights photovoltaic energy as an important resource and the most appealing one to meet today's energy requirements. The importance given to this kind of energy has attracted many researchers throughout the world to ensure the best benefit of sunlight. The interest was devoted to several axes concerning the development of new photovoltaic modules and achieving the most optimal way of their employment [5,6]. To that end, modeling photovoltaic generators and simulating their performance has also emerged as an important task to be done to facilitate and offer more opportunities to manipulate their device with lower costs [7,8]. Considering that achieving efficient and fast PV modeling allows the prediction of generated electricity, the estimation of the maximum output power under outdoor conditions, and makes the comparison of several PV array setups less expensive, modeling photovoltaic generators is considered a required task for performance assessment and the maximum power point tracking [9,10].

Photovoltaic modeling is founded on equivalent electrical circuits. The single-diode circuit model (SDM) and the double-diode circuit model (DDM), are the most adopted electrical models in the literature. The main parameters of the single-diode model are the light-generated current $I_L$, the shunt resistance $R_{sh}$, the series resistance $R_s$, the reverse saturation current $I_s$, and the ideality factor a [11]. For the double-diode model, in addition to the five parameters of the SDM another reverse saturation current $I_{s2}$, and ideality factor $a_2$ are added [12]. The wide

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**Nomenclature**

| | |
|---|---|
| a | Ideality factor for the SDM |
| $a_1$ | Ideality factor of $D_1$ for the DDM |
| $a_2$ | Ideality factor of $D_2$ for the DDM |
| AM | Air mass constant equals 1.5 |
| G ($W/m^2$) | Received irradiance |
| I (A) | Terminal current |
| $I_{cal}$ (A) | Calculated current |
| $I_L$(A) | Photo-generated current |
| $I_s$ (A) | Reverse saturation current for the SDM |
| $I_{Meas}$(A) | Measured current |
| $I_{mp}$(A) | Maximum power point's current |
| $I_{s1}$ (A) | Reverse saturation current of $D_1$ for the DDM |
| $I_{s2}$ (A) | Reverse saturation current of $D_2$ for the DDM |

*Abbreviation*

| | |
|---|---|
| AE | Absolute Error |
| AM | Air Mass |
| IJAYA | Improved JAYA Algorithm |
| IGSK | Improved Gaining-Sharing Knowledge Algorithm |
| DDM | Double-Diode Model |
| $K_B$(J/K) | Boltzmann constant equals $1.38 \times 10^{-23}$ J/K |
| $N_s$ | Number of PV module's cells in series |
| P (W) | Output power |
| $P_{C,max}$(W) | Calculated maximum power |
| $P_{M,max}$(W) | Measured maximum power |
| q (C) | Electron charge equals $1.6021 \times 10^{-19}$C |
| T(°C) | PV generator's temperature |
| $V_T$ (V) | Thermal voltage |
| $V_{mp}$ (V) | Maximum power point's voltage |
| $V_{oc}$ (V) | Open-circuit voltage |
| V (V) | Terminal voltage |
| $K_{Isc}$ (A/°C) | Temperature coefficient of $I_{sc}$ |
| MLBSA | Multiple Learning Backtracking Search Algorithm |
| MPP | Maximum Power Point |
| RMSE | Root Mean Square Error |
| SDM | Single-Diode Model |
| STC | Standard Test Conditions |

reliance on these two last models refers to their simplicity and modeling accuracy [13]. On the first hand, by neglecting the losses caused by the recombination phenomenon occurring in the depletion region, the one-diode equivalent circuit model avoids the calculation's complexity and keeps offering sufficient efficiency, as well, thus it is widely selected by the authors [14,15]. On the other hand, the two-diode equivalent circuit model that takes into account the recombination losses allows more modeling precision but raises the prediction challenge by increasing the number of unknown parameters to calculate from five to seven [16,17].

Based on several approaches, various methods differing in their degree of accuracy and complexity were proposed in the literature to predict PV generators' parameters and estimate their performance. The approaches introduced in the literature can be divided into two main categories, analytical and numerical techniques. Given the non-linearity nature of the current equation, the analytical methods are based on some assumptions to facilitate the calculations and guarantee sufficient precision of the estimation [18,19]. The numerical methods include the use of numerical resolution algorithms to solve a system of non-linear equations derived from the current equation, iterative approaches, and optimization techniques based on meta-heuristic algorithms and data fitting [20,21]. The major inconvenience of the numerical resolution is to find the initial values, especially for the saturation currents, the shunt resistance, and the ideality factors. The iterative methods find their limitation when the stopping constraints and the increment steps are not well chosen or when the number of loops increases. The optimization techniques generally require the availability of the experimental I–V characteristics, which limits the tasks devoted to the PV modeling, in addition to the requirement of selecting accurate values for the optimization parameters in the case of heuristic algorithms [22–25].

Kumar et al., have based on a purely numerical approach to propose a method applied for the SDM and DDM. The principle of this last is the extraction of five equations for the SDM and seven equations for the DDM based on many approximations to be solved using the "fsolve" function of MATLAB software. Then, derive five and seven equations using more approximations to get the required initial values for the numerical resolution [17]. However, the complete reliance on the numerical resolution of equations' systems, which all require initial guessed values, highly affects the convergence of the resolution and demands tedious calculations and more assumptions for simplification.

The authors in Ref. [26] have assumed that the series resistance equals zero and the shunt resistance is infinite. However, these lasts are valid only for the ideal solar cells, and in this case, the resistive losses are supposed to be totally absent. Thus, these assumptions are invalid when modeling real PV generators. Moreover, in Ref. [26], it has been accepted that the two reverse saturation currents of the DDM are connected by an approximate predetermined relation. Notwithstanding that the latter approximation does not have any physical significance or proof, it was first used by the authors in Ref. [27] and adopted later by several authors to simplify the needed calculations for the identification [28]. In addition to the approximation used for the two reverse saturation currents, in Ref. [28], the values of the two ideality factors were taken directly for each PV module from the two separate works in Refs. [29,30] which are based on different extraction approaches. The light-generated current's value was also extracted by Yahya-Khotbehsara et al., using an additional assumption admitting the $I_L$=$I_{sc}$ equality [28]. Then, after neglecting the series resistance versus the shunt resistance in some calculation stages for simplifications, it was extracted by the authors in Ref. [28] using an iterative process based on an equation adjustment technique proposed first by Majdoul et al., for the SDM [31].

In the attempt to identify the parameter's values of the DDM with a lower degree of complexity, Ishaque et al., [32] assume the $I_{s1} = I_{s2}$ equality, which was adopted later in several works [33,34]. However, the reverse saturation current is highly dependent on temperature level since it is a leakage current, while the second saturation current is derived from the second diode; thus, it models the electron-hole recombination current. Therefore, $I_{s2}$ takes higher values than $I_{s1}$, and assuming the equality of both currents has no physical sense [35]. In Refs. [32,33], the light-generated current is assumed to be equal to the short circuit current, and the first ideality factor is taken equal to 1 and the second ideality factor takes values greater than 1.2 directly assigned as fixed values to these two last parameters. For the works in Ref. [34], the two ideality factors are supposed to be equal and take values near 1. However, this approximation has no physical significance because the first ideality factor measures the resemblance between the diode's characteristics and those of an ideal diode, and it is generally close to 1 while, the second ideality factor corresponds to the recombination phenomenon and takes 2 as a value when the phenomenon dominates [35]. The series resistance and the shunt resistance are extracted in Refs. [32,33] using an iterative loop based on the maximum power point adjustment, the technique which was first used by Villalva et al., [36] for the SDM.

Zhuo et al., [37] have adopted the assumption admitting the equality of both saturation currents. Then, based on Shockley diffusion theory, 1

was assigned to the diode-ideal diode similarity factor and the value 2 to the recombination factor [38]. Next, the series resistance extraction went through tedious calculations and was done in the final step using an iterative process based on the mountain climbing algorithm. By contrast, recently, to avoid the fixed values used for the two ideality factors, Orioli et al., [39] and Chennoufi et al., [35] have chosen to increase the number of the used loops for the identification. Chennoufi et al., [35] have proposed two methods; in the first case, the two fixed values $a_1 = 1$ and $a_2 = 2$ have been used. Then, the $R_s$ value was extracted using the Brent method. In the second case, to improve the results, two successive iterative processes were used to extract the two ideality factors' values and the series resistance based on error minimization at the MPP. As a result, the iterative process provides almost 1 for the first ideality factor and 2 for the second [6,35]. In this case, the use of two separate loops highly affects the required execution time for the estimation [6]. Orioli et al., [39] have proposed an iterative technique based on three sequent loops to get the values of $R_s$, $a_1$, and $a_2$, successively. Indeed, increasing the number of iterative processes may lead to more accurate results but amplifies the convergence time, which can be considered a negative point for real-time applications [6].

To extract the values of the electrical parameters, latterly, several optimization techniques employing the heuristic algorithms were also introduced in the literature, for example, the use of the JAYA and improved JAYA (IJAYA) algorithms [40], the multiple learning backtracking search algorithm (MLBSA) [41], and the improved gaining-sharing knowledge algorithm (IGSK) [42]. However, these kinds of optimization techniques are time-consuming due to their counting on huge stochastic populations. Moreover, they require having preliminary knowledge about the variation interval of each parameter to extract, the thing that cannot always be available. Furthermore, whenever these algorithms are not well initialized, or if the intervals of variation were not properly chosen, they become unable to converge and produce satisfactory results. In addition to the selection of initial values for the model's parameters, these last algorithms require also a good predefinition of their own linked parameters, which varies from one algorithm to another [43].

The objective of this paper is to propose a hybrid prediction technique that is adaptable to both DDM and SDM models for estimating the performance of PV generators. The ideality factors' values are first determined using the Shockley diffusion theory [44]. According to this theory, the diode-ideal diode similarity factor is equal to 1 and the recombination ideality factor's value is 2. Then, the proposed method uses only the values of the I–V curve's remarkable points to compute the other parameters. In fact, the identification methodology is divided into two extraction approaches using the I–V equation at the short circuit current, the maximum power point (MPP), and the open-circuit voltage, in addition to the use of the current's derivative with respect to the voltage (V) at the MPP. The first approach is an analytical method based on matrix resolution, which is used to calculate the parameters that are not obvious to be initialized, including the saturation currents ($I_s$ for the SDM and $I_{s1}$ and $I_{s2}$ for the DDM) and the light-generated currents of both electrical models. The second method is the numerical approach based on a fast iterative process to calculate the $R_{sh}$ and $R_s$ values for both DDM and SDM. The extraction of the shunt resistance is done by using a new slope adjustment technique at the linear zone of the I–V curve. Moreover, to guarantee high rapidity of convergence of the slope searching towards the accurate solution, the search interval is defined with great precision around the approximate value of $R_{sh}$. The increment steps are also defined and generalized using a new method devoted to selecting the increment steps relative to the initially found value of $R_{sh}$, by taking into consideration the small error that can be committed when extracting the approximate value of $R_{sh}$. The starting value is determined based on the I–V curve's slope calculation at the linear zone. Then, by utilizing the found value of $R_{sh}$, $R_s$ is calculated using the Brent hybrid root-finding method [45].

The method in the current work is applied to different PV generators of various technologies, including mono-crystalline technology, poly-crystalline technology, and thin film technology. The effectiveness test for the proposed method is done based on the calculation of some statistical indicators such as the RMSE (Root Mean Square Error), as well as the convergence speed. Then, the obtained values of these last indicators are evaluated against some of the well-known and most reviewed methods in this field.

In an attempt to organize the present paper, the SDM and the DDM are detailed in section 2, the proposed method's theory is introduced in section 3, section 4 is devoted to the results and the test of the effectiveness, and section 5 gives the conclusion of the works.

## 2. Equivalent circuit of PV modules

Photovoltaic modules are constructed from the combination of identical solar cells arranged in series and parallel to hit the required currents and voltages. To model a single solar cell, two models are selected in this work. The first model shown in Fig. 1 is the one-diode model in which the used diode goes back to the juxtaposition of two semiconductors of different doping nature inside the solar cell, and it gives rise to two electrical parameters; the ideality factor a and the reverse saturation current $I_s$. The model consist also of the generator of current that models the light-generated current to which the current $I_L$ parameter is associated. The two last electrical parameters are $R_s$ which is the series resistance that models the losses due to the joule effect, and the shunt resistance $R_{sh}$ modeling the leaks through the PV cell's PN-junction related to the imperfection of the used materials [20,23].

The non-linear mathematical equation linking the terminal current I of a PV generator to the terminal voltage V is extracted based on the single-diode model using the mesh law applied to the circuit of Fig. 1, and it is written as follows [46]:

$$I = I_L - I_s\left[\exp\left(\frac{IR_s + V}{aV_T N_s}\right) - 1\right] - G_{sh}(IR_s + V) \quad (1)$$

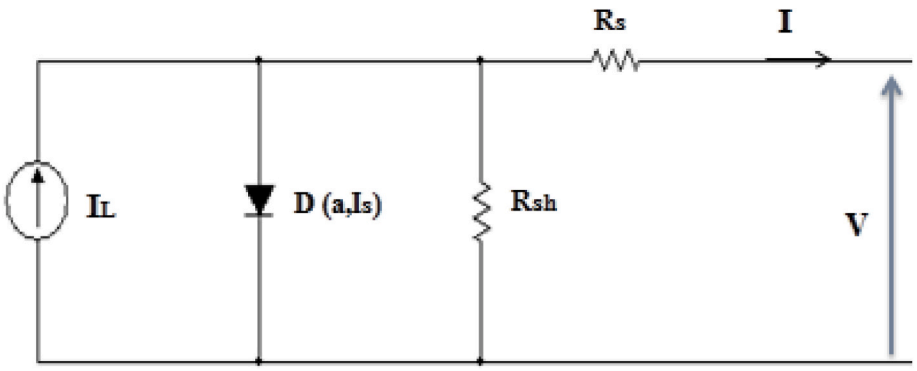



**Fig. 1.** Single-diode model.

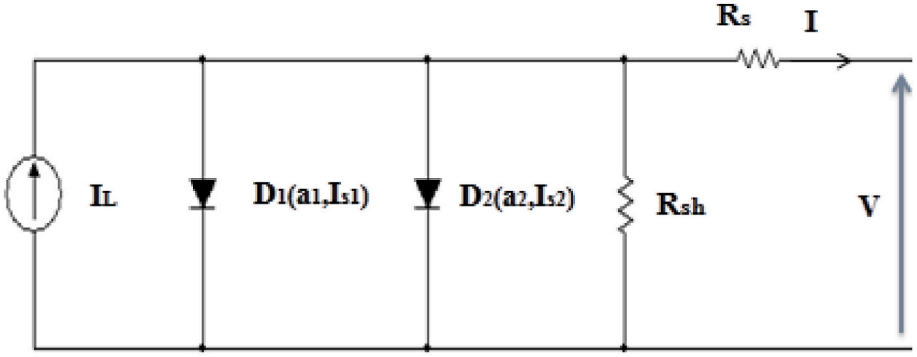



**Fig. 2.** Double-diode model.

where $N_s$ is the number of series cells of the PV generator, the thermal voltage $V_T$ is given as $V_T=(K_B/q) \times T$, where $K_B$ is the Boltzmann constant, q is the electron charge, and T is the PV cell's back surface temperature. $G_{sh}$ is the shunt admittance defined as $G_{sh} = 1/R_{sh}$.

The second selected electrical model is the double-diode model shown in Fig. 2. In addition to the effects considered in the SDM, the DDM takes into consideration the recombination phenomenon occurring in the PV cell's space charge region. This last effect is modeled by a second diode added to the SDM to form the DDM with two more parameters. Hence, the two-diode model contains seven unknown parameters to be determined ($I_L$, $I_{s1}$, $I_{s2}$, $a_1$, $a_2$, $R_s$, and $R_{sh}$), where $a_1$ and $I_{s1}$ are, respectively, the ideality factor and the reverse saturation current corresponding to the first diode, and $a_2$ and $I_{s2}$ the ideality factor and the reverse saturation current corresponding to second diode [27, 28].

Based on the double-diode model, the current equation of a PV generator is extracted using the mesh law applied to the circuit in Fig. 2, and it is written as follows:

$$I = I_L - I_{s1}\left[\exp\left(\frac{IR_s + V}{a_1 V_T N_s}\right) - 1\right] - I_{s2}\left[\exp\left(\frac{IR_s + V}{a_2 V_T N_s}\right) - 1\right] - G_{sh}(IR_s + V) \tag{2}$$

## 3. Parameter extraction strategies

To simulate PV generators' performance, this section introduces an extraction technique to first calculate the unknown parameters, which are the important keys to PV simulation. The proposed technique is characterized by its flexibility, so that it can be adopted for both PV models; the DDM and the SDM. The issue that parameter extraction often confronts is the problem of negative values which have no physical significance. To avoid this last issue, the calculation methodology is divided into two parts. The first part is the analytical calculation of the light-generated current and the reverse saturation currents. The second part contains the adjustment of the slope in the vicinity of the short-circuit current to extract the shunt resistance and the numerical extraction of the series resistance.

### 3.1. Analytical methodology

#### 3.1.1. Double-diode model

For the DDM, to get the values of $I_L$, $I_{s1}$, and $I_{s2}$, analytically, the current equation gotten from the DDM is evaluated at the mean points of the I–V characteristic and written as follows:

For ($I=I_{sc}$ and $V = 0$)

$$I_{sc} = I_L - I_{s1}\left[\exp\left(\frac{I_{sc}R_s}{a_1 V_T N_s}\right) - 1\right] - I_{s2}\left[\exp\left(\frac{I_{sc}R_s}{a_2 V_T N_s}\right) - 1\right] - G_{sh}I_{sc}R_s \tag{3}$$

For ($I=I_{mp}$ and $V=V_{mp}$)

$$I_{mp} = I_L - I_{s1}\left[\exp\left(\frac{I_{mp}R_s + V_{mp}}{a_1 V_T N_s}\right) - 1\right] - I_{s2}\left[\exp\left(\frac{I_{mp}R_s + V_{mp}}{a_2 V_T N_s}\right) - 1\right] - G_{sh}(I_{mp}R_s + V_{mp}) \tag{4}$$

For ($I = 0$ and $V=V_{oc}$)

$$I_L - I_{s1}\left[\exp\left(\frac{V_{oc}}{a_1 V_T N_s}\right) - 1\right] - I_{s2}\left[\exp\left(\frac{V_{oc}}{a_2 V_T N_s}\right) - 1\right] - V_{oc}G_{sh} = 0 \tag{5}$$

By adopting the notations of Eq. (6) below:

$$C_{sc1,2} = \exp\left(\frac{I_{sc}R_s}{a_{1,2}V_T N_s}\right) - 1 \quad ; \quad C_{mp1,2} = \exp\left(\frac{I_{mp}R_s + V_{mp}}{a_{1,2}V_T N_s}\right) - 1 \; ; \quad C_{oc1,2} = \exp\left(\frac{V_{oc}}{a_{1,2}V_T N_s}\right) - 1 \tag{6}$$

Eqn 3, 4 and 5 can be managed and combined into a system of equations, which is equivalent to the equation (7).

$$\begin{cases} I_L - C_{sc,1}I_{s1} - C_{sc,2}I_{s2} = I_{sc} + \dfrac{I_{sc}R_s}{R_{sh}} \\ I_L - C_{oc,1}I_{s1} - C_{oc,2}I_{s2} = \dfrac{V_{oc}}{R_{sh}} \\ I_L - C_{mp,1}I_{s1} - C_{mp,2}I_{s2} = I_{mp} + \dfrac{(I_{mp}R_s + V_{mp})}{R_{sh}} \end{cases} \tag{7}$$

The system of Eq. (7) can be rearranged and written as follows:

$$A_{DDM} \times \lambda_{DDM} = \beta_{DDM} \tag{8}$$

where:

$$A_{DDM} = \begin{pmatrix} 1 & -C_{sc,1} & -C_{sc,2} \\ 1 & -C_{oc,1} & -C_{oc,2} \\ 1 & -C_{mp,1} & -C_{mp,2} \end{pmatrix} \tag{9}$$

$$\beta_{DDM} = \begin{pmatrix} I_{sc} + \dfrac{I_{sc}R_s}{R_{sh}} \\ \dfrac{V_{oc}}{R_{sh}} \\ I_{mp} + \dfrac{(I_{mp}R_s + V_{mp})}{R_{sh}} \end{pmatrix} \tag{10}$$

$$\lambda_{DDM} = \begin{pmatrix} I_L(R_s) \\ I_{s1}(R_s) \\ I_{s2}(R_s) \end{pmatrix} \tag{11}$$

The Eq. (12) below is solved to get the values of the three parameters of the DDM; $I_L$, $I_{s1}$, and $I_{s2}$.

$$\lambda_{DDM} = A_{DDM}^{-1} \times \beta_{DDM} \tag{12}$$

### 3.1.2. Single-diode model

For the SDM the same analytical procedure introduced for the DDM is being used to extract $I_L$ and $I_s$. At the short circuit and the maximum power points of the I–V characteristics, the current equation based on the SDM is written as follows [20]:

$$I_{sc} = I_L - I_s\left[\exp\left(\frac{I_{sc}R_s}{aV_TN_s}\right) - 1\right] - G_{sh}I_{sc}R_s \tag{13}$$

$$I_{mp} = I_L - I_s\left[\exp\left(\frac{I_{mp}R_s + V_{mp}}{aV_TN_s}\right) - 1\right] - G_{sh}\left(I_{mp}R_s + V_{mp}\right) \tag{14}$$

To simplify the equations' writing, the following notations are adopted.

$$\begin{aligned} &C_{sc} = \exp\left(\frac{I_{sc}R_s}{aV_TN_s}\right) - 1 \quad ; \quad C_{mp} = \exp\left(\frac{I_{mp}R_s + V_{mp}}{aV_TN_s}\right) - 1 \; ; \\ &C_{oc} = \exp\left(\frac{V_{oc}}{aV_TN_s}\right) - 1 \end{aligned} \tag{15}$$

The two equations' system is obtained by combining the Eq. (13) and the Eq. (14).

$$\begin{cases} I_L - C_{sc}I_s = I_{sc} + \dfrac{I_{sc}R_s}{R_{sh}} \\ I_L - C_{mp}I_s = I_{mp} + \dfrac{\left(I_{mp}R_s + V_{mp}\right)}{R_{sh}} \end{cases} \tag{16}$$

The system of Eq. (16) above is transformed into the matrix equation below:

$$A_{SDM} \times \lambda_{SDM} = \beta_{SDM} \tag{17}$$

where A is the two array matrix defined as:

$$A_{SDM} = \begin{pmatrix} 1 & -C_{sc} \\ 1 & -C_{mp} \end{pmatrix} \tag{18}$$

and β and λ are two vectors defined as:

$$\beta_{SDM} = \begin{pmatrix} I_{sc} + \dfrac{I_{sc}R_s}{R_{sh}} \\ I_{mp} + \dfrac{\left(I_{mp}R_s + V_{mp}\right)}{R_{sh}} \end{pmatrix} \tag{19}$$

$$\lambda_{SDM} = \begin{pmatrix} I_L(R_s) \\ I_s(R_s) \end{pmatrix} \tag{20}$$

To get the values of the two parameters $I_L$ and $I_s$ the Eq. (21) is used.

$$\lambda_{SDM} = A_{SDM}^{-1} \times \beta_{SDM} \tag{21}$$

## 3.2. Numerical methodology to extract $R_s$ and $R_{sh}$

In this section, the steps to be followed to extract the shunt resistance and the series resistance's values based on the DDM and the SDM using only the key-points of the current-voltage characteristic are introduced. To that end, the extraction strategy is divided into two parts. The first is the calculation of $R_{sh}$ using an iterative process. The second is the use of the extracted value of $R_{sh}$ to get the $R_s$ value by solving a nonlinear equation using Brent's hybrid root-finding method [45].

### 3.2.1. Double-diode model

This subsection is dedicated to introducing the proposed technique for extracting $R_{sh}$ and $R_s$ values in the context of DDM. The technique involves solving the derivative equation of the current at the maximum power point (MPP) using the Brent method. To do this, the derivative of the current's equation of Eq. (2) is computed with respect to the voltage V, resulting in the following expression [17]:

$$\begin{aligned} \frac{dI}{dV} = &-\frac{I_{s1}}{a_1V_TN_s}\left(R_s\frac{dI}{dV} + 1\right) \times \exp\left(\frac{IR_s + V}{a_1V_TN_s}\right) - \frac{I_{s2}}{a_2V_TN_s}\left(R_s\frac{dI}{dV} + 1\right) \\ &\times \exp\left(\frac{IR_s + V}{a_2V_TN_s}\right) - \frac{1}{R_{sh}}\left(R_s\frac{dI}{dV} + 1\right) \end{aligned} \tag{22}$$

At the maximum power point (MPP), the derivative of the power equation with respect to the voltage (V) is equal to zero.

$$\left.\frac{dP}{dV}\right|_{(P=P_{mp})} = 0 \tag{23}$$

However, in order to calculate the derivative of the output power with respect to the output voltage (V) using the Eq. (22), it is necessary to employ the equation (24) below, which gives the expression of P.

$$P = I \times V \tag{24}$$

Thus, the derivative of the power expression with respect to the voltage (V) can be found by deriving the Eq. (24) with respect to V, leading to the resulting Eq. (25):

$$\frac{dP}{dV} = V \times \frac{dI}{dV} + I \tag{25}$$

By combining the Eq. (23) and the Eq. (25), the current's derivative equation with respect to the voltage evaluated at the MPP can be obtained as given in the Eq. (26) below:

$$\left.\frac{dI}{dV}\right|_{(I=I_{mp},V=V_{mp})} = -\frac{I_{mp}}{V_{mp}} \tag{26}$$

By evaluating the Eq. (22) at the MPP and using the expression of the Eq. (26), the Eq. (27) below can be obtained. The value of $R_s$ can be then determined by solving this equation using the Brent method.

$$\begin{aligned} &\frac{-I_{mp}}{V_{mp}} + \frac{I_{s1}}{a_1V_TN_s}\left(1 - \frac{I_{mp}R_s}{V_{mp}}\right) \times \left(C_{mp,1} - 1\right) + \frac{I_{s2}}{a_2V_TN_s}\left(1 - \frac{I_{mp}R_s}{V_{mp}}\right) \\ &\times \left(C_{mp,2} - 1\right) + \frac{1}{R_{sh}} \times \left(1 - \frac{I_{mp}R_s}{V_{mp}}\right) = 0 \end{aligned} \tag{27}$$

$R_{sh}$ is still the only unknown parameter in Eq. (27) to be determined before solving the equation. For the purpose of extracting some of the unknown parameters, many authors propose relying only on iterations and a stopping criterion [35,39]. The technique is commonly used in the literature to get the $R_s$ values because it is the easiest parameter to be initialized as it takes small values, so near to zero [28,36,37]. However, for some PV modules, the series resistance takes values near 2, in this case, the wrong choice of the increment steps can lead to the requirement of a large number of iterations, and that impacts the convergence time. In addition to $R_s$, several authors proceed likewise to get the values of the ideality factors, as well [35,39]. The last technique further threatens the appropriateness of the execution time [6]. Therefore, to reduce the number of iterations and guarantees the reproduction of the algorithm, the increment steps must be well-determined and adapted to each PV generator of any technology. To that end, it is not obvious to find an increment step's value for $R_s$ because of its small values and its appearance in both exponential terms of the current equation of the DDM in addition to its influence on the third term. Hence, the three I–V curve's zones are affected by this last sensitive parameter. As a solution, and also to avoid the probable negative values of $R_{sh}$, a new technic based on the adjustment of the shunt resistance's value is proposed in the current work. Indeed, the method consists of choosing the research interval of $R_{sh}$ around its initial value extracted by calculating the slope's value at the vicinity of the short circuit point or the theoretical calculation of the initial value of $R_{sh}$. The principle consists in solving the equation (28) below:

$$\left.\frac{dI}{dV}\right|_{I=Isc,V=0} = -\frac{1}{R_{sh0}} \tag{28}$$

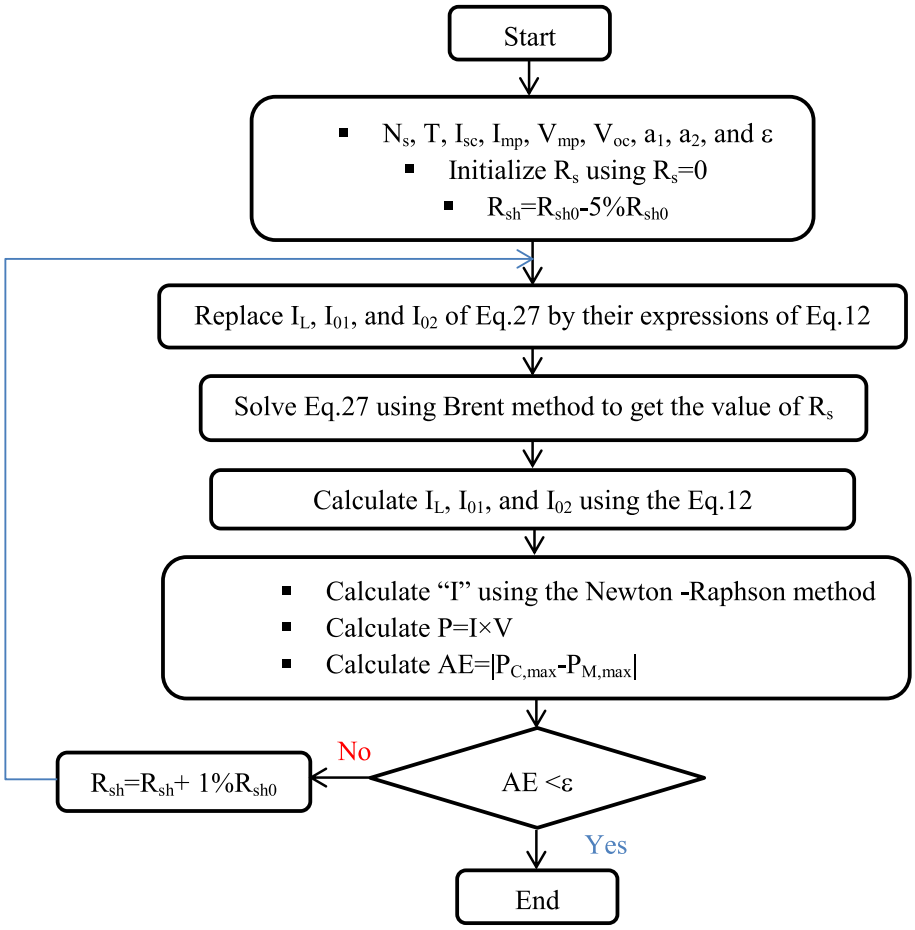



**Fig. 3.** Flow chart for the DDM.

The supplementary benefit of the proposed technique for $R_{sh}$ is its flexibility so that it can be adapted to the case of data availability, in this case, the Eq. (28) can be solved graphically. Otherwise, the theoretical initial value of $R_{sh}$ given below can be used [36]:

$$R_{sh0} = \frac{V_{mp}}{I_{sc} - I_{mp}} - \frac{V_{oc} - V_{mp}}{I_{mp}} \tag{29}$$

The searching interval is defined with consideration of the visible 5% of error around the initial value, to ensure selecting another initial value of $R_{sh}$, which corresponds to a slope below the desired slope that will match the most effective value of $R_{sh}$. Then, the stopping criterion consists in minimizing the absolute error between the calculated MPP value ($P_{C,max}$) and the measured one ($P_{M,max}$) given in Eq. (30), until the desired precision corresponding to the tolerance (ε) is reached.

$$AE(W) = \left|P_{C,max} - P_{M,max}\right| \tag{30}$$

The flow chart of the proposed method based on the DDM is given in Fig. 3.

### 3.2.2. Single-diode model

To make the numerical technique adapted with the SDM to extract the value of $R_s$, the Eq. (31) obtained from the calculation of the current equation's derivative with respect to the voltage (V) at the MPP [17,47], is solved using Brent's method.

$$\frac{-I_{mp}}{V_{mp}} + \frac{I_s}{aV_T N_s}\left(1 - \frac{I_{mp}R_s}{V_{mp}}\right) \times \left(C_{mp} - 1\right) + \frac{1}{R_{sh}} \times \left(1 - \frac{I_{mp}R_s}{V_{mp}}\right) = 0 \tag{31}$$

The value of $R_{sh}$ is extracted by following the same numerical method proposed for the DDM.

The flow chart for the SDM is given in Fig. 4.

### 3.3. Estimating the performance of PV modules at various weather conditions

Calculating the electrical parameters of PV generators at the standard test conditions (T = 25 °C, G = 1 kW/m$^2$, and AM = 1.5) is considered mandatory to achieve the estimation of PV modules' performance when operating under external conditions. However, the calculated parameters at STC are affected by weather variations, including temperature and irradiance. Thus, the transfer of parameters' values from STC to non-STC is a needed task. Accordingly, the transfer equations depending on T and G levels are widely used in the literature to meet that need [6,35,37].

The ideality factors of the DDM are presumed unaffected neither by the irradiance nor temperature variations [6,35].

$$a_1(T, G) = a_{1,ref} \tag{32}$$

$$a_2(T, G) = a_{2,ref} \tag{33}$$

The shunt resistance and the series resistance's values vary only when the level of irradiance varies, and they are linked to the G level by the following formulas [6]:

$$R_{sh,ref}(T, G) = R_{sh,ref}\frac{G_{ref}}{G} \tag{34}$$

$$R_s(T, G) = R_{s,ref}\frac{G_{ref}}{G} \tag{35}$$

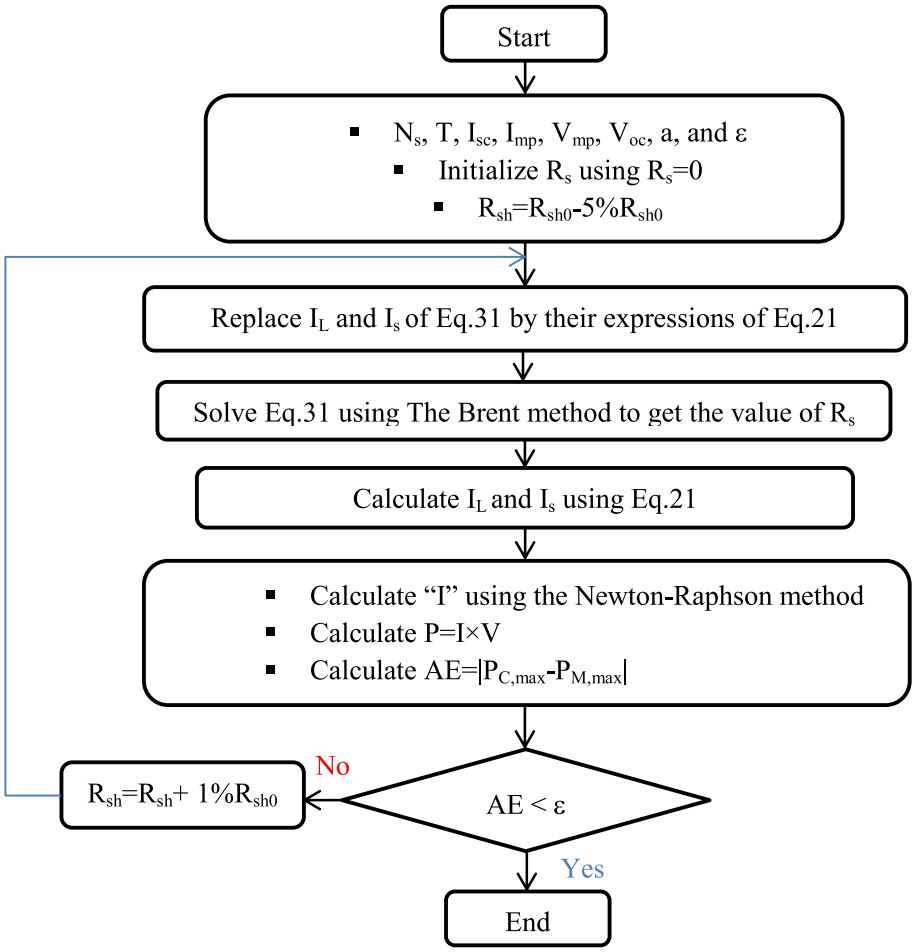



**Fig. 4.** Flow chart for the SDM.

The values taken by $I_{s1}$ and $I_{s2}$ are influenced by the temperature variations and can be determined at non-STC using $I_{s1,ref}$ and $I_{s2,ref}$ values calculated at the STC, the equations allowing this transformation are given below [35,37]:

$$I_{s1}(T,G) = I_{s1,ref} \times \exp\left(\frac{T}{V_T a_{1,ref}}\left(\frac{E_{g,ref}}{T_{ref}} - \frac{E_g}{T}\right)\right) \times \left(\frac{T}{T_{ref}}\right)^3 \quad (36)$$

$$I_{s2}(T,G) = I_{s2,ref} \times \exp\left(\frac{T}{V_T a_{2,ref}}\left(\frac{E_{g,ref}}{T_{ref}} - \frac{E_g}{T}\right)\right) \times \left(\frac{T}{T_{ref}}\right)^3 \quad (37)$$

where $E_g$ notation represents the band gap energy at external conditions defined as [31,35].

$$E_g = E_{g,ref}\left(1 + 0.0002677 \times \left(T_{ref} - T\right)\right) \quad (38)$$

The light-generated current is affected by both temperature and irradiance variations according to the following equation:

$$I_{ph}(T,G) = I_{ph,ref}\left(1 - K_{Isc}\left(T_{ref} - T\right)\right)\left(\frac{G}{G_{ref}}\right) \quad (39)$$

The index "ref" is used to designate the parameter's values at the reference conditions (STC).

## 4. Results and discussion

This section is dedicated to validating the effectiveness of the proposed method. In the literature, numerous PV modules have been extensively analyzed in the PV modeling field. Generally, these modules represent the widely installed technologies such as poly-crystalline, mono-crystalline, and thin film technologies. Among them, mono-crystalline PV systems are considered the most efficient and are well-suited for warm and sunny locations. While polycrystalline modules may have slightly lower performance than mono-crystalline ones, they offer a competitive cost-efficiency ratio [48]. On the other hand, as reported by the authors in Ref. [49], thin film technology is characterized by a lower temperature coefficients, maintaining stable performance even at higher temperatures. Accordingly, this technology is characterized by high performance ratio. Thus, thin film modules represent an excellent choice for hotter climates, despite their lower efficiency [48,49].

In this work, to comprehensively and meaningfully evaluate the new approach, the validation of the modeling technique includes six widely used PV modules from previous technologies [6,32,37]. The mean points' values at Standard Test Conditions (STC) corresponding to each PV module and their required temperature coefficients are provided in Table 1.

For a fair comparison, at first, the electrical parameters of the equivalent circuit models are calculated under STC (G = 1 kW/m$^2$, T = 25 °C, and AM = 1.5) and then compared with other proposed approaches in the literature. Next, the I–V characteristics are estimated based on the extracted parameters and tested against the measured data of the PV modules using some statistical indicators. The resulting values of this last comparison are also compared with the results provided by some of the most reviewed approaches in the literature. During the second phase, the achieved results at STC are used to prediction at non-STC.

**Table 1**
PV modules' specific parameters.

| | Poly-crystalline | | | Mono-crystalline | | Thin film |
|---|---|---|---|---|---|---|
| | BP solar MSX-60 | KD245GH-4FB2 | S75 | SQ150-PC | SM55 | ST40 |
| $N_s$ | 36 | 60 | 36 | 72 | 36 | 36 |
| $I_{sc}$ (A) | 3.8 | 8.91 | 4.7 | 4.80 | 3.45 | 2.68 |
| $I_{mp}$(A) | 3.5 | 8.23 | 4.26 | 4.40 | 3.15 | 2.41 |
| $V_{mp}$(V) | 17.1 | 29.8 | 17.6 | 34 | 17.4 | 16.6 |
| $V_{oc}$(V) | 21.1 | 36.9 | 21.6 | 43.4 | 21.7 | 23.3 |
| $K_{Isc}$ (A/°C) | $3 \times 10^{-3}$ | $5.35 \times 10^{-3}$ | $2 \times 10^{-3}$ | $1.4 \times 10^{-3}$ | $1.4 \times 10^{-3}$ | $3.5 \times 10^{-4}$ |

### *4.1. Effect of the shunt resistance's adjustment on the I–V curves' prediction and results for STC*

The current work introduces several new extraction techniques. Thus, this subsection is laid aside first for the visualization of the shunt resistance's adjustment effect on the estimation of the PV generators' characteristics. Second, to test the impact of eliminating the two parameters related to the second diode of the DDM to return to the SDM on the proposed method's performance. The third task to which this subsection is devoted is the test of the proposed approach's effectiveness at the STC by validating the accuracy against the methods introduced in the literature. And then finally, to visualize the predicted I–V curves at the reference conditions.

#### *4.1.1. Effect of shunt resistance's adjustment*

The test in this part is done over the multi-crystalline PV panel BP solar MSX-60 operating at STC. Fig. 5 provides a clear visualization of the effect of adjusting the $R_{sh}$ value on the slope adjustment at the short circuit point and also its effect on the slope of the I–V curve's first zone for the BP solar MSX-60 PV module. As Fig. 5(a) shows, when $R_{sh}$ takes its least value it implies the highest slope near $I_{sc}$. Then, when $R_{sh}$ increases, the inclination decreases to reach its final value that matches the MPP and then, fits the linear zone with great precision. The previous observation is enhanced by Fig. 5(b) giving the individual absolute errors calculated between the measured data and the calculated characteristic for each $R_{sh}$ value. It can be seen that by adjusting $R_{sh}$, the absolute error minimizes for the great majority of the predicted curve.

Table 2 gives the calculated values of the DDM parameters using the slope adjustment technique in comparison with three of the widely reviewed works given in Refs. [17,37], and [50]. The root mean square error (RMSE) is chosen as the statistical indicator to measure the resemblance ratio between the measured and calculated characteristics based on the extracted DDM parameters. This choice also enables an accurate comparison with previous techniques. According to the results, the new modeling approach demonstrates the best similitude to the measured data, with an RMSE value of 0.0343 A, which is the least value among the other methods. The method in Ref. [17] is a purely numerical method based on a resolution of seven equations using the 'fsolve' function of MATLAB software. However, the technique is based on several approximations to simplify the tedious calculation, in addition to

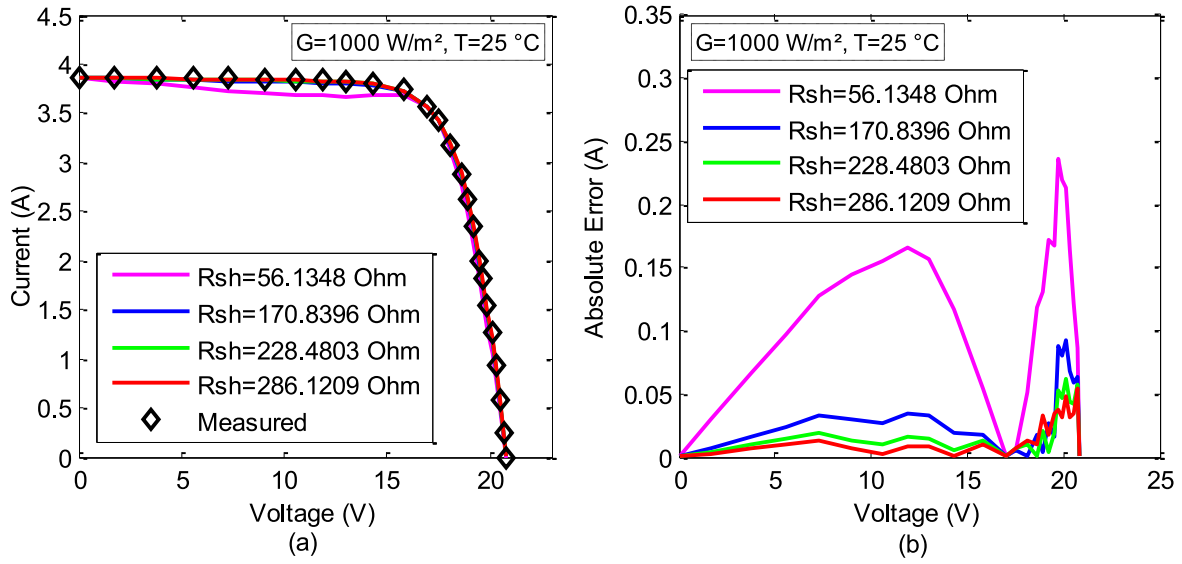



**Fig. 5.** (a) Estimated characteristic for MSX-60 PV module for various $R_{sh}$ values compared to the measured characteristic. **(b)** Absolute errors matching each $R_{sh}$ value.

**Table 2**
Extracted DDM's parameters for the MSX-60 PV module at STC using the $R_{sh}$ adjustment technic compared to three other modeling methods of literature.

| | BP solar MSX-60 | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | $I_L$(A) | $a_1$ | $a_2$ | $I_{s1}$(A) | $I_{s2}$(A) | $R_{sh}$(Ω) | $R_s$(Ω) | RMSE (A) |
| **Kumar et al.**, [17] | 3.8046 | 0.99859 | 2.0014 | $3.9901 \times 10^{-10}$ | $4.033 \times 10^{-06}$ | 280.2171 | 0.3397 | 0.1908 |
| **Zhuo et al.**, [37] | 3.808361 | 1 | 2 | $4.597752 \times 10^{-10}$ | $4.597752 \times 10^{-10}$ | 169.081308 | 0.372046 | 0.1666 |
| **Elbaset et al.**, [50] | 3.8084 | 1.0003 | 1.9997 | $4.8723 \times 10^{-10}$ | $6.1528 \times 10^{-10}$ | 169.0471 | 0.3692 | 0.1379 |
| **Proposed DDM Method** | 3.8618 | 1 | 2 | $6.0842 \times 10^{-10}$ | $3.1092 \times 10^{-06}$ | 286.1209 | 0.2904 | 0.0343 |

the use of seven initializations. These factors have led to the least precision of estimation by providing 0.1908 A as an RMSE value, which is the highest value. Moreover, it can be seen that the values of the two ideality factors consistently return toward 1 for $a_1$ and 2 for $a_2$, this observation is also confirmed by the results provided by the works in Ref. [50]. The last observation trend aligns with the physical significance of these two parameters.

#### 4.1.2. Predicting PV modules performance at STC using the modeling DDM and SDM methods

Intending to test the effectiveness of the new modeling approach applied for both the DDM and the SDM against the existing numerical methods in the literature, the multi-crystalline S75 and the mono-crystalline SM55 PV panels operating at reference conditions are used in this part. Table 3 compares the DDM's parameters and SDM's five parameters extracted using the proposed modeling technique applied to the multi-crystalline S75 PV panel, also compared to six other works introduced in the literature. Table 4 gives the same study done over the mono-crystalline SM55 PV. For a relevant evaluation, the new modeling DDM method is also tested over the multi-crystalline KD245GH-4FB2, the mono-crystalline SQ150-PC PV module, and the thin-film ST40 PV module, and its efficiency is compared with three different methods of literature. The results are shown, respectively, in Tables 5–7.

As appears in Tables 3 and 4 for the multi-crystalline S75 and the mono-crystalline SM55 the highest estimation's accuracy among the other predicting techniques is offered by the proposed modeling method applied for the DDM by providing 0.0334 A as RMSE value for S75 PV module and 0.0364 A for the SM55 PV module. Moreover, when the adjusting technique is applied to the SDM, it provides high levels of precision compared with the existing works in the literature. If comparing the obtained DDM's parameters with the SDM's ones, it can be observed that for both PV modules, the SDM provides almost the same values of $I_L$, $I_{s1}$, $a_1$, and $R_s$ as those yielded by the DDM. However, it can be observed from the difference between the achieved $R_{sh}$ values that when the approach is used for the SDM, it converges relatively faster than the DDM, as the SDM gives the lowest values of this parameter. However, since $R_{sh}$ takes high values and is incremented based on a percentage technique, the convergence time is not heavily influenced by the small differences in $R_{sh}$ values. Furthermore, the small difference between the values of $R_{sh}$ of the two models and the consideration of the existence of the recombination phenomenon give the DDM the preference at the precision level. So it can be concluded that even if the DDM requires more steps to converge, it achieves the most accurate results. The absolute neglect of the resistive effects inside PV generators by the authors in Ref. [26] is the reason for obtaining less accurate perdition by this technique for both PV modules. While, the high values of RMSE obtained by the methods in Refs. [32,33] for the two PV panels are explained by the use of a series of approximations including $I_L$=$I_{sc}$ and $I_{s1} = I_{s2}$ equalities that have no physical explanation. The method in Ref. [29] is based on a meta-heuristic technique affected by the additional parameters related to the used algorithm, and that influences as well the estimation's precision. The mean disadvantage of the works in

**Table 3**
Extracted DDM and SDM's parameters for the multi-crystalline S75 PV module at STC using the adjustment technique compared to three other modeling methods of literature.

| | Shell S75 | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | $I_L$(A) | $a_1$ | $a_2$ | $I_{s1}$(A) | $I_{s2}$(A) | $R_{sh}$(Ω) | $R_s$(Ω) | RMSE (A) |
| **Tifidat et al.,** [6] | 4.70163 | 1.09371 | 1.9420 | $2.114 \times 10^{-09}$ | $4.24173 \times 10^{-06}$ | 756.458 | 0.2612 | 0.075788 |
| **Chitti Babu et al.,** [26] | 4.7 | 1.82 | 3.36 | $1.23 \times 10^{-05}$ | $3.18 \times 10^{-05}$ | ∞ | 0 | 0.222448 |
| **Yahya-Khotbehsara et al.,** [28] | 4.7 | 1.14 | 2.6 | $5.68 \times 10^{-09}$ | $1.47 \times 10^{-08}$ | 103.58 | 0.23 | 0.121921 |
| **Ishaque et al.,** [32] | 4.7 | 1 | 1.3 | $3.39 \times 10^{-10}$ | $3.39 \times 10^{-10}$ | 84.38 | 0.27 | 0.089657 |
| **Khezzar et al.,** [33] | 4.7 | 1 | 1.3 | $3.388 \times 10^{-10}$ | $3.388 \times 10^{-10}$ | 78.641 | 0.2541 | 0.107319 |
| **Chin et al.,** [29] | 4.7 | 1.14 | 2.60 | $1.63 \times 10^{-08}$ | $9.29 \times 10^{-06}$ | 403.16 | 0.25 | 1.147528 |
| **Proposed Method for SDM** | 4.70987 | 1 | – | $3.2976 \times 10^{-10}$ | – | 159.011 | 0.33399 | 0.038862 |
| **Proposed Method for DDM** | 4.70936 | 1 | 2 | $3.3026 \times 10^{-10}$ | $1.4034 \times 10^{-08}$ | 170.507 | 0.33975 | 0.033388 |

**Table 4**
Extracted DDM and SDM's parameters for the Mono-crystalline SM55 PV module at STC using the adjustment technique compared to three other modeling methods of literature.

| | Shell SM55 | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | $I_L$(A) | $a_1$ | $a_2$ | $I_{s1}$(A) | $I_{s2}$(A) | $R_{sh}$(Ω) | $R_s$(Ω) | RMSE(A) |
| **Chitti Babu et al.,** [26] | 3.45 | 1.89 | 3.65 | $1.4300 \times 10^{-05}$ | $3.7000 \times 10^{-05}$ | ∞ | 0 | 0.1062 |
| **Yahya-Khotbehsara et al.,** [28] | 3.45 | 1.26 | 2.84 | $2.7500 \times 10^{-08}$ | $7.1100 \times 10^{-08}$ | 233.4600 | 0.36000 | 0.0789 |
| **Khezzar et al.,** [33] | 3.45 | 1 | 1.3 | $2.4873 \times 10^{-10}$ | $2.4873 \times 10^{-10}$ | 93.03299 | 0.29890 | 0.1211 |
| **Lun et al.,** [34] | 3.4619 | 1.01658 | 1.01658 | $1.5529 \times 10^{-10}$ | $1.5524 \times 10^{-10}$ | 144.2891 | 0.49696 | 0.0577 |
| **Chennoufi et al.,** [35] | 3.4604 | 0.98580 | 2.01420 | $1.4249 \times 10^{-10}$ | $2.1336 \times 10^{-06}$ | 151.3306 | 0.45670 | 0.0642 |
| **Zhuo et al.,** [37] | 3.4624 | 1 | 2 | $2.1593 \times 10^{-10}$ | $2.1593 \times 10^{-10}$ | 141.2190 | 0.50639 | 0.0478 |
| **Proposed Method for SDM** | 3.4621 | 1 | – | $2.3781 \times 10^{-10}$ | – | 132.1152 | 0.46150 | 0.0393 |
| **Proposed Method for DDM** | 3.4616 | 1 | 2 | $2.3812 \times 10^{-10}$ | $4.1270 \times 10^{-08}$ | 140.3952 | 0.47230 | 0.0364 |

**Table 5**
Extracted DDM's parameters for the multi-crystalline KD245GH-4FB2 PV module at STC using the adjustment technique compared to three other modeling methods of literature.

| | Keyocera KD245GH-4FB2 | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | $I_L$(A) | $a_1$ | $a_2$ | $I_{s1}$(A) | $I_{s2}$(A) | $R_{sh}$(Ω) | $R_s$(Ω) | RMSE (A) |
| **Enebish et al.,** [51] | 8.9335 | 1.0004 | 2.0008 | $3.5748 \times 10^{-10}$ | $-1.1878 \times 10^{-6}$ | 120.1507 | 0.3163 | 0.0475 |
| **Kumar et al.,** [17] | 8.9202 | 0.9954 | 2.0044 | $2.7637 \times 10^{-10}$ | $6.4751 \times 10^{-6}$ | 247.9658 | 0.2834 | 0.07356 |
| **Orioli et al.,** [39] | 8.9337 | 0.9735 | 1.5649 | $1.8084 \times 10^{-10}$ | $1.9062 \times 10^{-10}$ | 120.1628 | 0.3192 | 0.04855 |
| **Proposed Method for DDM** | 8.9310 | 1 | 2 | $3.4743 \times 10^{-10}$ | $6.6148 \times 10^{-8}$ | 131.5809 | 0.3103 | 0.0444 |

**Table 6**
Extracted DDM's parameters for the mono-crystalline SQ150-PC PV module at STC using the adjustment technique compared to three other modeling methods of literature.

| | Shell SQ150-PC | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | $I_L$(A) | $a_1$ | $a_2$ | $I_{s1}$(A) | $I_{s2}$(A) | $R_{sh}$(Ω) | $R_s$(Ω) | RMSE (A) |
| **Khezzar et al.,** [33] | 4.8 | 1 | 1.3 | $3.1068 \times 10^{-10}$ | $3.1068 \times 10^{-10}$ | 275 | 0.9 | 0.04299 |
| **Chitti Babu et al.,** [26] | 4.8 | 2.0300 | ∞ | $4.5836 \times 10^{-05}$ | $1.1875 \times 10^{-04}$ | ∞ | 0 | 0.27081 |
| **Orioli et al.,** [39] | 4.8054 | 0.73076 | 1.47759 | $1.8765 \times 10^{-19}$ | $5.9857 \times 10^{-07}$ | 444.1415 | 0.4981 | 0.12066 |
| **Proposed DDM Method** | 4.8092 | 1 | 2 | $2.8725 \times 10^{-10}$ | $4.1830 \times 10^{-06}$ | 444.6400 | 0.8503 | 0.0294 |

**Table 7**
Extracted DDM's parameters for the thin-film ST40 PV module at STC using the adjustment technique compared to three other modeling methods of literature.

| | Shell ST40 | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | $I_L$(A) | $a_1$ | $a_2$ | $I_{s1}$(A) | $I_{s2}$(A) | $R_{sh}$(Ω) | $R_s$(Ω) | RMSE (A) |
| **Yahya-Khotbehsara et al.,** [28] | 2.68 | 1.52 | 2.15 | $1.65 \times 10^{-07}$ | $4.28 \times 10^{-07}$ | 1350.30 | 1.40 | 0.06393 |
| **Ishaque et al.,** [32] | 2.68 | 1 | 1.3 | $3.07 \times 10^{-11}$ | $3.07 \times 10^{-11}$ | 198.94 | 1.71 | 0.0957 |
| **Zhuo et al.,** [37] | 2.70773 | 1 | 2 | $2.9755 \times 10^{-11}$ | $2.97551 \times 10^{-11}$ | 168.9173 | 1.74749 | 0.094321 |
| **Proposed DDM Method** | 2.7115 | 1 | 2 | $2.2134 \times 10^{-11}$ | $1.2153 \times 10^{-06}$ | 127.6626 | 1.5184 | 0.0319 |

Ref. [34] that may explain the obtained RMSE value for the SM55 PV module is the equality between the two ideality factors and the two saturation currents' equality.

According to the results in Table 5, the first observation to be made is that the proposed adjustment technique yields the lowest value of RMSE for KD245GH-4FB2 PV when compared to other methods of the literature. The second note is that although the works in Refs. [17,51] use two different techniques, they both converge toward 1 for $a_1$ and 2 for $a_2$. The results of Table 6 confirm the effectiveness of the proposed method which yields the least values of the selected statistical indicators also for the mono-crystalline SQ150-PC PV module. Furthermore, the used approximations in Ref. [26] have supplied a physical unexplained value of the second ideality factor for the SQ150-PC, and that has also led to an increase of the corresponding RMSE value of this method. The same observation concerning the method's accuracy can be noted from Table 7 for the thin-film ST40 PV module. The proposed methods in Refs. [32,37] assume the equality of the two reverse saturation currents and that has strongly affected the efficiency of the prediction.

Figs. 6(a) and 7(a) show the estimated I–V characteristics using the proposed adjustment approach applied to the DDM tested over the two PV modules; the shell S75 and the shell SM55, respectively, operating at the STC. The predicted curves are presented by the continuous red line, the methods of literature are given by the colored lines, and the measured current-voltage curves are displayed using black markers. For both PV modules, it can be seen that in the vicinity of the MPP, the introduced technique in the current paper ensures high prediction accuracy, and the estimated characteristics match with high precision the experimental data. Figs. 6(b) and 7(b) give the individual absolute errors matching each of the used methods for the extraction of I–V characteristics of the two PV modules S75 and SM55, respectively. The calculated absolute errors at the vicinity of the MPP enhance the previous observation on the proposed method's effectiveness for the MPP's prediction.

With the aim of testing the speed of convergence of the proposed method, this last, as well as the three methods in Refs. [26,28], and [32], were implemented in MATLAB R2012ab software and have been run several times on an hp 8470p EliteBook with 4 GB RAM and an Intel Core i5-3320M CPU, 2.60 GHz processor. The provided computational time and errors at the MPP for S75 and SM55 PV modules are shown in Table 8. As reported in the table, the best estimation of MPP is provided by the proposed adjustment technique at the least execution time for both PV modules compared to the other methods. These two last results are achieved first due to the use of absolute error minimization at the

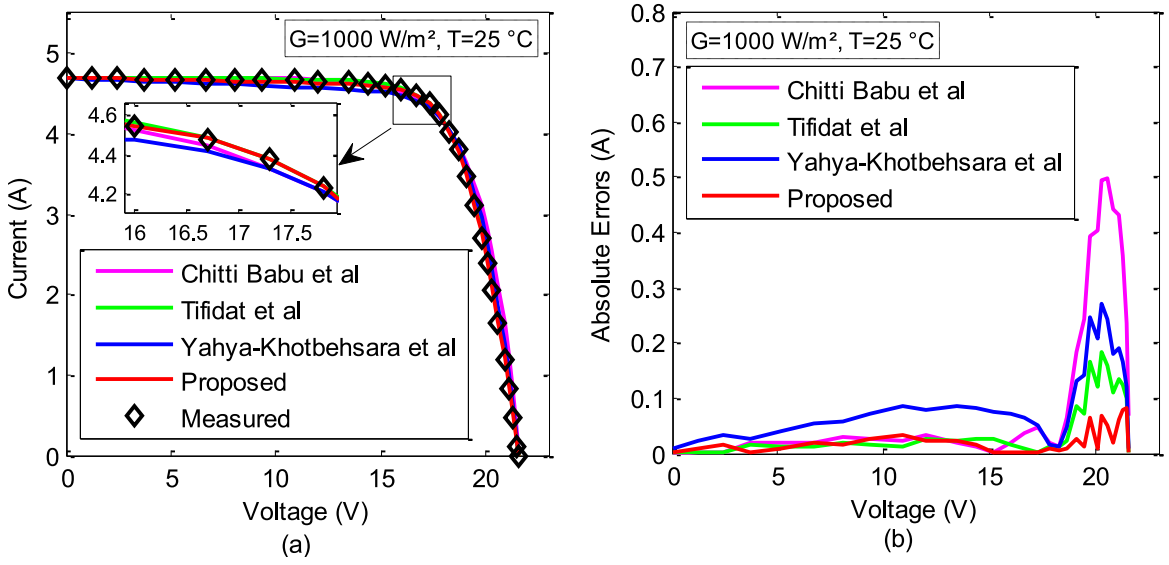



**Fig. 6.** (a) Estimated characteristic for the multi-crystalline S75 PV module at STC compared with the measured data and other methods of the literature. (**b**) Individual absolute errors matching each modeling method.

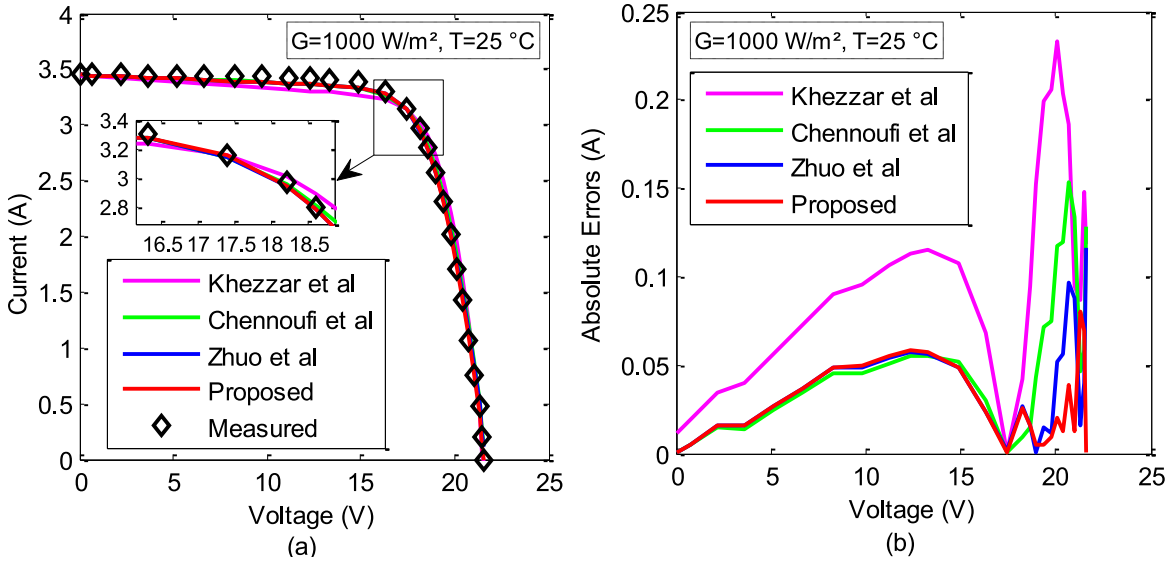



**Fig. 7.** (a) Estimated characteristic for the mono-crystalline SM55 PV module at STC compared with the measured data and other methods of the literature. (**b**) Individual absolute errors matching each modeling method.

**Table 8**
Comparison of the proposed model's convergence time and accuracy for MPP's prediction with existing modeling techniques.

| | | Chitti Babu et al., [26] | Yahya-Khotbehsara et al., [28] | Ishaque et al., [32] | Proposed DDM Method |
|---|---|---|---|---|---|
| **Poly-crystalline KD245GH-4FB2** | $P_{M,max}$ (W) | 245.2540 | 245.2540 | 245.2540 | 245.2540 |
| | $P_{C,max}$ (W) | 245.2483 | 245.2533 | 245.2540 | 245.2540 |
| | $AE_{Mpp}$ | 0.0057 | 0.0007 | $3.9999 \times 10^{-06}$ | $7.136397 \times 10^{-07}$ |
| | t (s) | 0.3164 | 0.1057 | 0.0910 | 0.083505 |
| **Mono-crystalline SM55** | $P_{M,max}$ (W) | 54.8100 | 54.8100 | 54.8100 | 54.8100 |
| | $P_{C,max}$ (W) | 54.6876 | 54.7885 | 54.8129 | 54.8100 |
| | $AE_{Mpp}$ | 0.1224 | 0.0215 | 0.0029 | $1.0869 \times 10^{-06}$ |
| | t (s) | 1.3022 | 0.2213 | 0.0887 | 0.074438 |

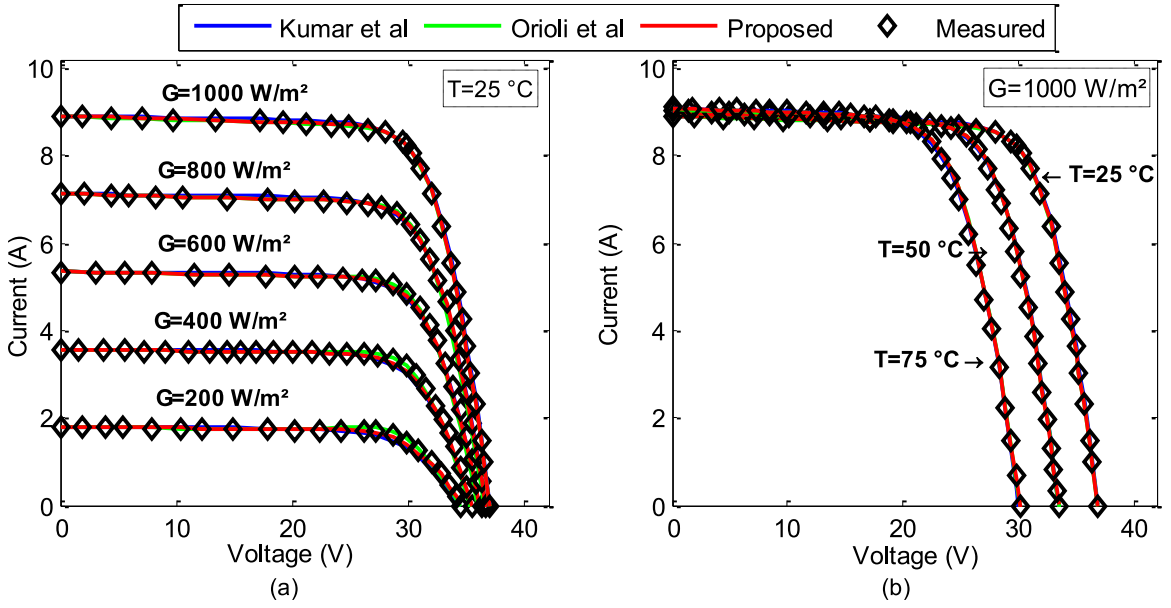



**Fig. 8.** (a) Estimated characteristic for the KD245GH-4FB2 PV module operating under various levels of illumination and standard T compared with the measured data and other methods of the literature. (**b**) Estimated and measured curves for the KD245GH-4FB2 PV module operating under various temperatures and standard G.

MPP and second due to the exploitation of the new technique allowing the fast extraction of the shunt resistance based on the increment using the percentage method. By contrast, the other methods of literature are based on random increments of $R_s$ using randomly selected steps chosen by the authors and applied for all PV modules, the fact which explains the high execution times of these methods.

### 4.2. Predicting PV modules performance at non-STC using the modeling DDM method

For the purpose of evaluating the performance of the modeling technique at non-STC, the DDM's parameters are utilized to estimate the PV generators' performance at external conditions by transferring their values to non-STC and calculating the I–V characteristics. Subsequently, the correspondence of the predicted performance is tested against the measured I–V curves, and the results are compared to other methods reported in the literature. This study is done over three PV modules, The multi-crystalline KD245GH-4FB2, the mono-crystalline SQ150-PC, and the thin-film ST40 PV panels. Fig. 8(a), Fig. 9(a), and Fig. 10(a) show the calculated and measured curves under the reference temperature and various G levels for the three PV modules, respectively. While, Figs. 8(b), 9(b) and 10(b) give the calculated and measured curves under the reference irradiance and various T levels for the same PV generators. From the figures, as the calculated curves highly match the measured data for all PV generators used for the test, it can be concluded that the new technique allows the estimation of PV performance under different weather conditions with great precision.

The modeling methods' RMSE related to each operating value of G and T = 25 °C calculated for the KD245GH-4FB2, the SQ150-PC, and the ST40 PV panels are given in Fig. 11(a), Fig. 12(a), and Fig. 13(a), respectively. It can be noted from the three figures that the new adjustment technique offers a high level of precision and follows the

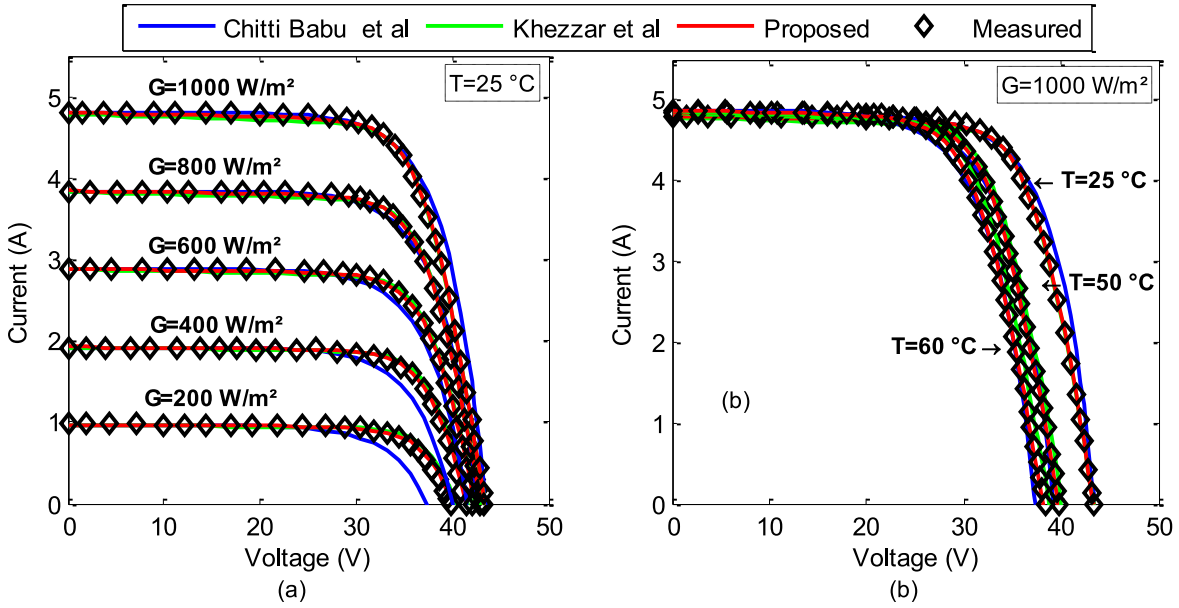



**Fig. 9.** (a) Estimated characteristic for the SQ150-PC PV module operating under various levels of illumination and standard T compared with the measured data and other methods of the literature. (**b**) Estimated and measured curves for the SQ150-PC PV module operating under various temperatures and standard G.

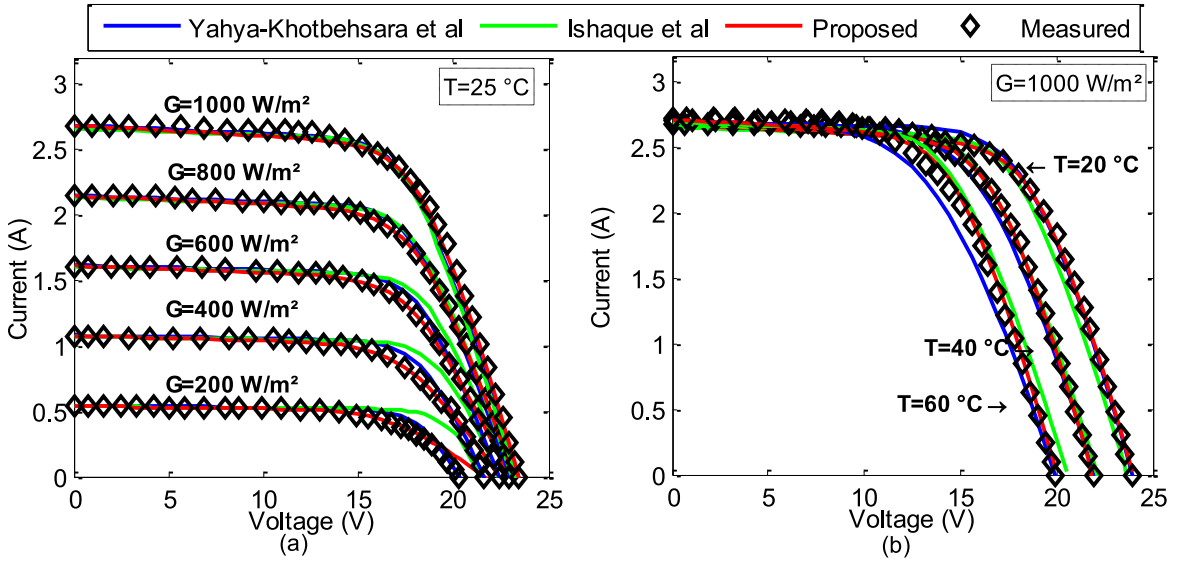



**Fig. 10.** (a) Estimated characteristic for the ST40 PV module operating under various levels of illumination and standard T compared with the measured data and other methods of literature. (**b**) Estimated and measured curves for the ST40 PV module operating under various temperatures and standard G.

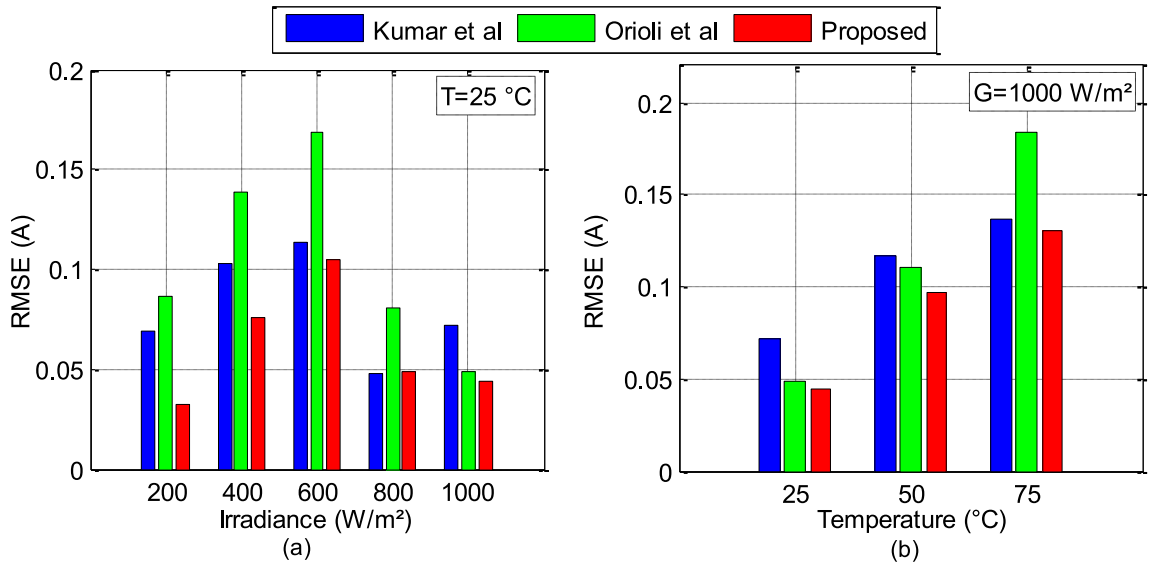



**Fig. 11.** (a) RMSE of each modeling method applied for the KD245GH-4FB2 module operating under various levels of illumination and standard T. (**b**) RMSE of each modeling method applied for the KD245GH-4FB2 PV module operating under various temperatures and standard G.

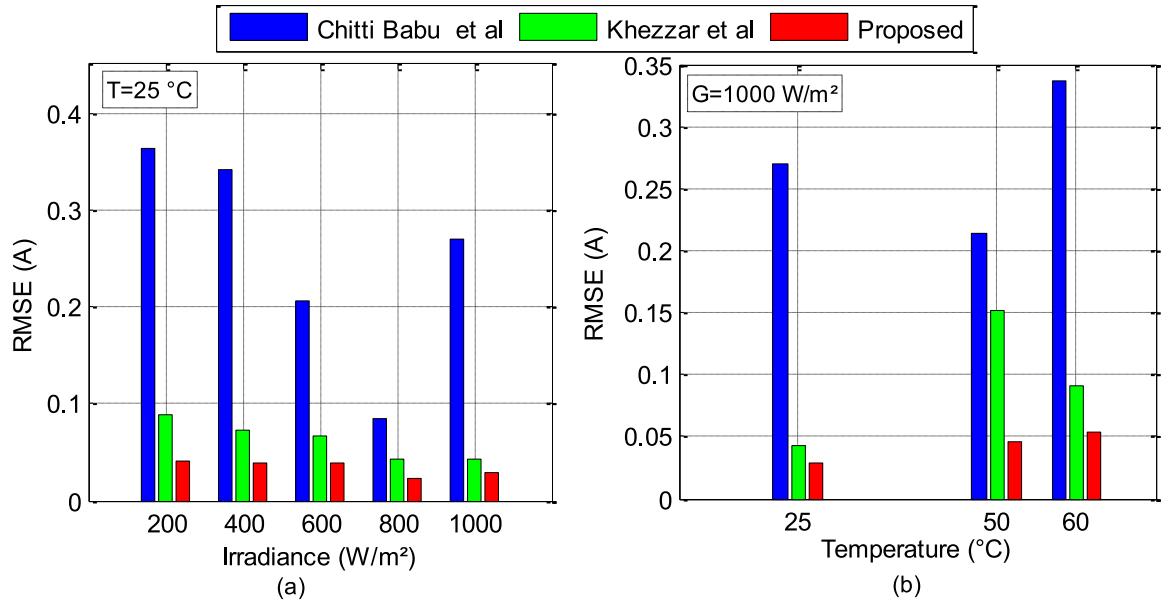



**Fig. 12.** (a) RMSE of each modeling method applied for the SQ150-PC PV module operating under various levels of illumination and standard T. (**b**) RMSE of each modeling method applied for the SQ150-PC PV module operating under various temperatures and standard G.

illumination's variations with great accuracy, and that is concluded from the RMSE that does not exceed 0.104 A for the multi-crystalline, 0.0411 A for the mono-crystalline, and 0.0466 A for the thin-film. On the other side, for the three technologies, respectively, the works in Ref. [39] reach 0.169 A, while 0.364 A is achieved by the method in Ref. [26] and 0.132 A by the technique in Ref. [32]. Figs. 11(b), 12(b) and 13(b) show the methods' RMSE related to each operating value of T and G = 1 kW/m$^2$, calculated for the same PV modules, respectively. The proposed parameter extraction technique provides in the worst cases RMSE = 0.13 A for the multi-crystalline, RMSE = 0.0542 A for the mono-crystalline, and RMSE = 0.0505 A for the thin-film, which are the least values provided when compared with the available methods in the literature. Consequently, it can be noted that the method in this work follows the weather variation with great accuracy when predicting I–V characteristics.

## 5. Conclusion

A fast modeling technique to calculate the parameters of the two-diode model of PV modules based on the slope adjustment using only the key-points' values at STC is introduced in the current paper. The method consists of extracting the three parameters; $I_L$, $I_{s1}$, and $I_{s2}$ using the analytical resolution of a three equation's system. The series resistance is extracted using an accurate numerical resolution based on the Brent's solver algorithm, and the shunt resistance is calculated based on a new fast adjusting technique. The modeling approach was also adapted to the single-diode model. The accuracy and the speed of convergence are evaluated against a considerable number of the available methods in the literature using six modules of various PV technologies when operating STC and non-STC conditions. The test of the efficiency was done using some statistical indicators for which the proposed technique gives the best values in the least execution time,

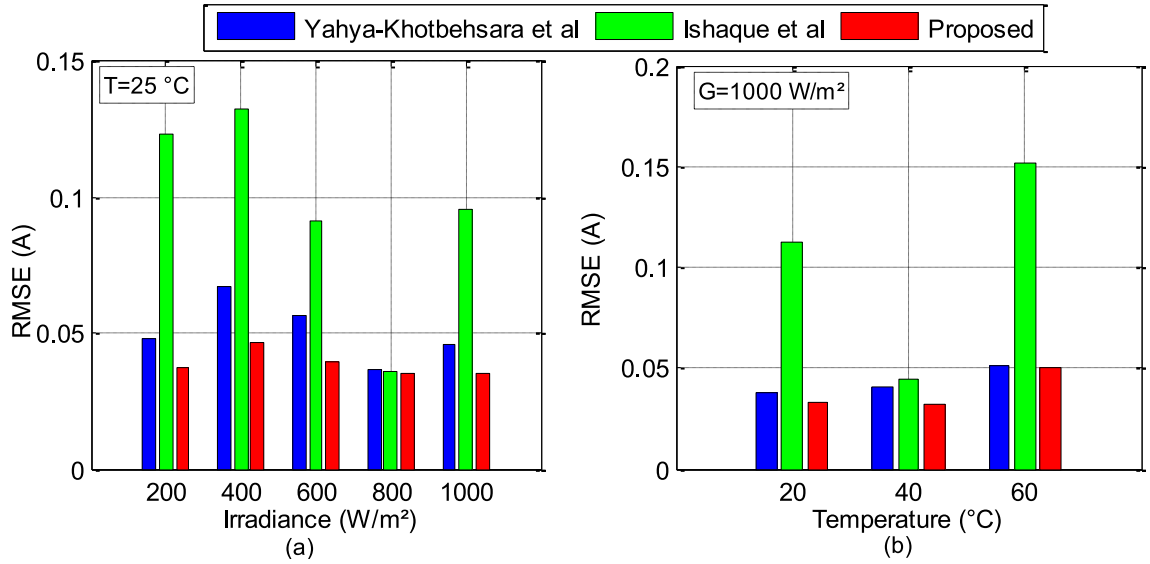



**Fig. 13.** (a) RMSE of each modeling method applied for the ST40 PV module operating under various levels of illumination and standard T. (**b**) RMSE of each modeling method applied for the ST40 PV module operating under various temperatures and standard G.

which makes the new method applicable for performance assessment at real-time and also for MPP tracking applications.

## CRediT authorship contribution statement

**Kawtar Tifidat:** Conceptualization, Methodology, Formal analysis, Software, Validation, Visualization, Investigation, Writing – original draft, Writing – review & editing. **Noureddine Maouhoub:** Conceptualization, Methodology, Validation, Investigation, Supervision, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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